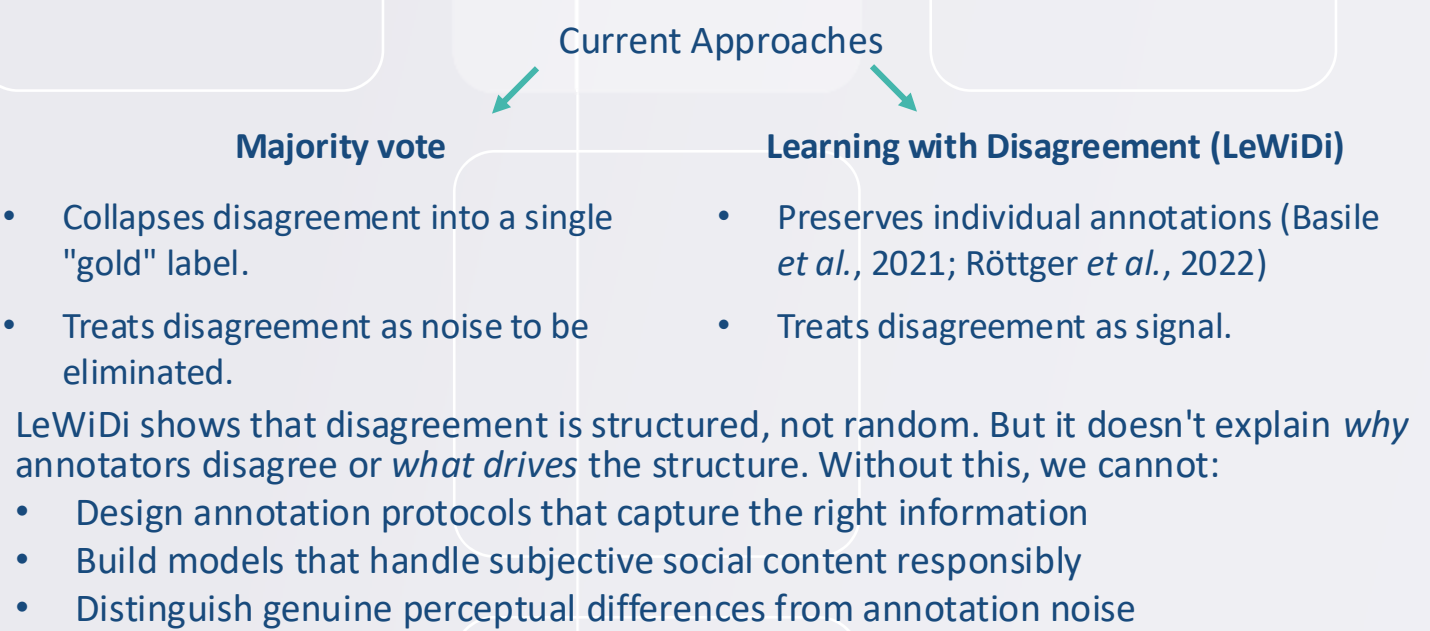
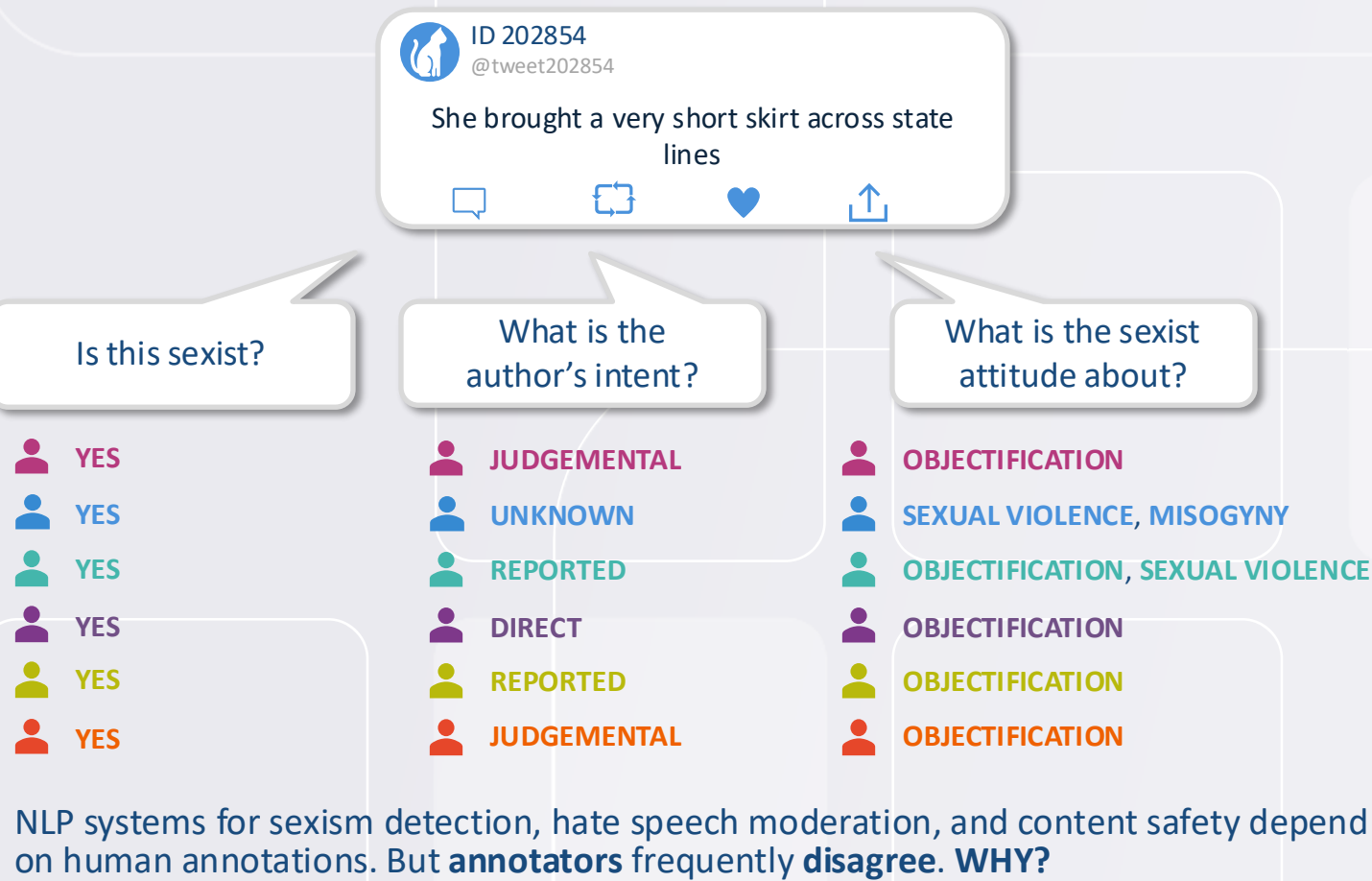


Social Perception as Theory of Mind: Evidence from Annotator Disagreement in Sexism Detection

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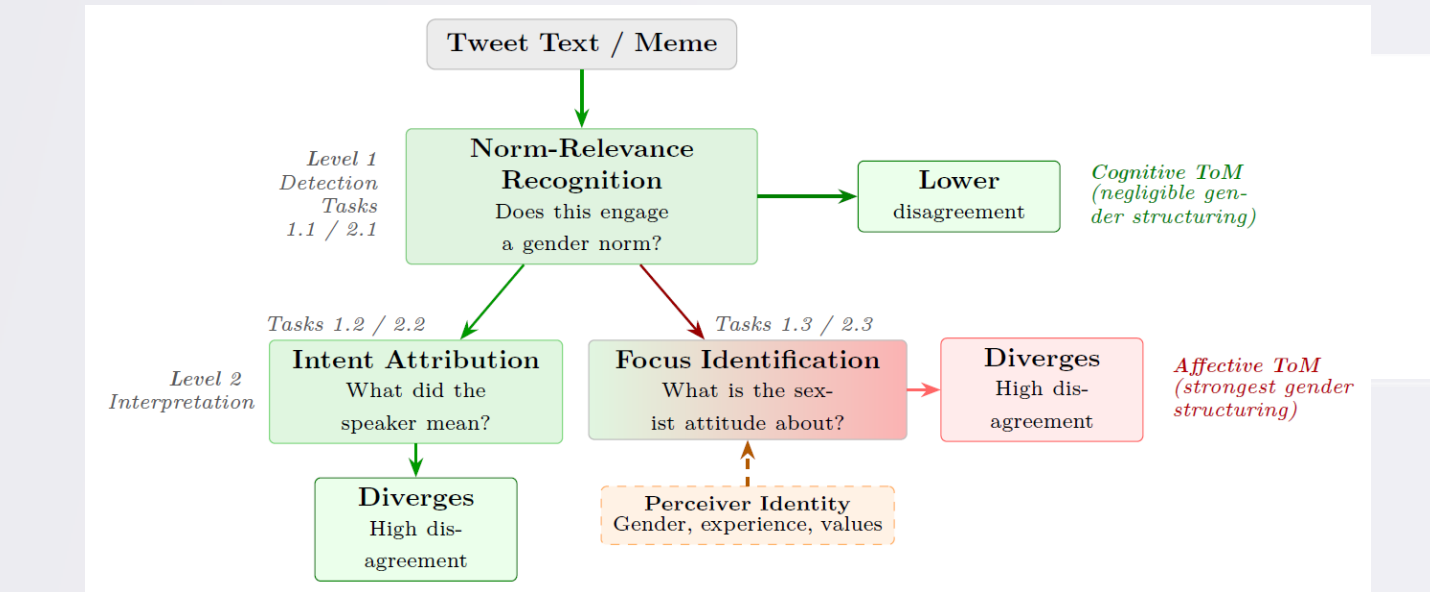
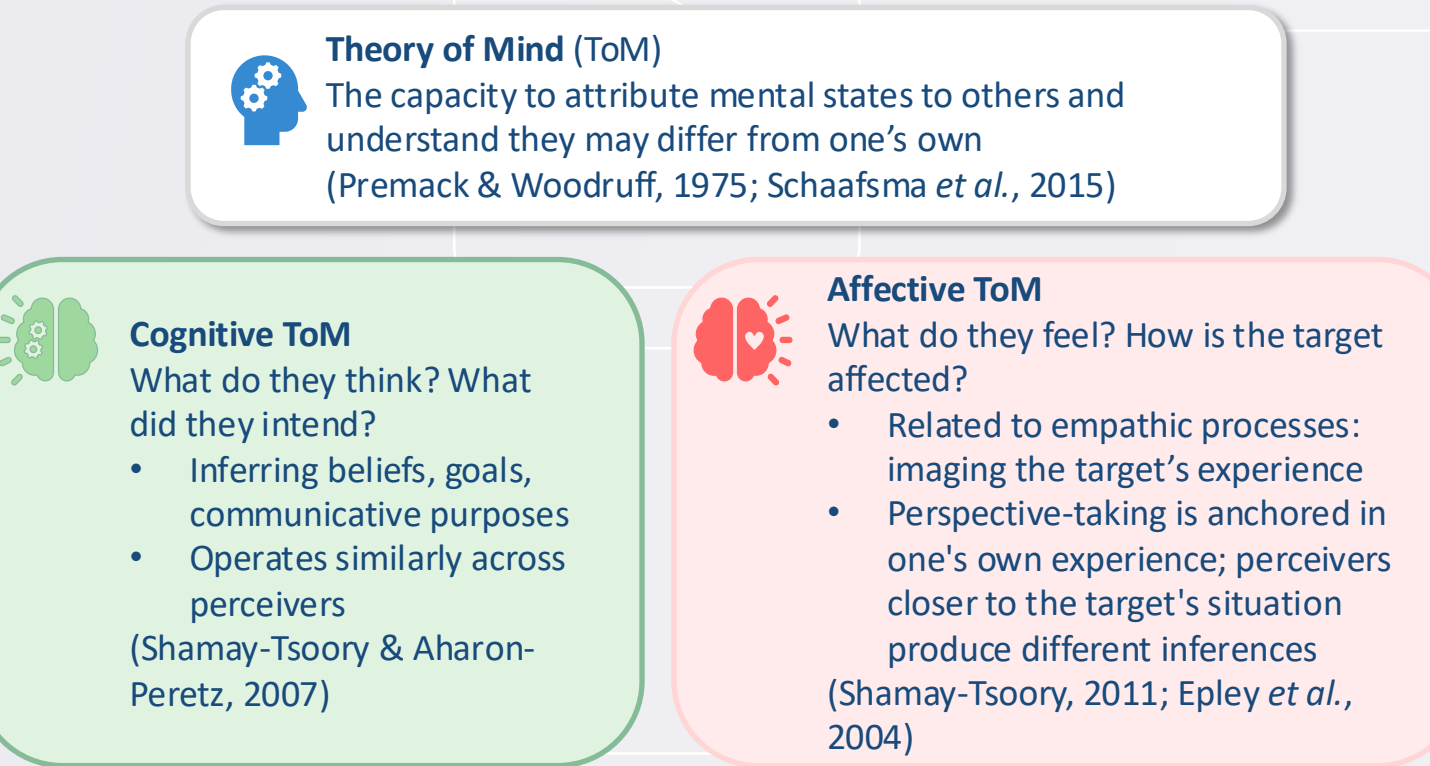
Motivation



Our proposal: *Theory of Mind (ToM)* provides this explanatory account:

- Different annotation tasks engage different cognitive processes.
- Perceiver identity selectively shapes the processes that require simulating others' experiences.

ToM Model



Cognitive ToM is a prerequisite for *affective ToM* (lesion evidence: Shamay-Tsoory & Aharon-Peretz, 2007). But *affective ToM* additionally depends on the perceiver's own experience → annotator identity matters selectively.

References

- Basile, V. et al. (2021). We Need to Consider Disagreement in Evaluation. *Proc. BPPF Workshop*.
- Becker, J. C. & Swim, J. K. (2011). Seeing the unseen: Attention to daily encounters with sexism. *Psych. Women Q.* 35(2).
- Epley, N. et al. (2004). Perspective taking as egocentric anchoring and adjustment. *JSPS* 87(3).
- Fredrickson, B. L. & Roberts, T.-A. (1997). Objectification theory. *Psych. Women Q.* 21(2).
- Glick, P. & Fiske, S. T. (1996). The Ambivalent Sexism Inventory. *JSPS* 70(3).
- Leonardelli, E. et al. (2023). SemEval-2023 Task 11: Learning with Disagreements. *Proc. SemEval*.
- Mulvey, L. (1975). Visual pleasure and narrative cinema. *Screen* 16(3).
- Perner, J. & Wimmer, H. (1985). Second-order belief attribution. *J. Exp. Child Psych.* 39(3).
- Plaza, L. et al. (2025). Overview of EXIST 2025. *CLEF 2025, LNCS* 16089.
- Premack, D. & Woodruff, G. (1978). Does the chimpanzee have a theory of mind? *BBS* 1(4).
- Röttger, P. et al. (2022). Two contrasting data annotation paradigms. *Proc. NAACL*.
- Schaafsma, S. M. et al. (2015). Deconstructing and reconstructing theory of mind. *TICS* 19(2).
- Shamay-Tsoory, S. G. & Aharon-Peretz, J. (2007). Dissociable prefrontal networks for cognitive and affective ToM. *Neuropsychologia* 45(13).
- Shamay-Tsoory, S. G. (2011). The neural bases for empathy. *The Neuroscientist* 17(1).
- Swim, J. K. & Cohen, L. L. (1997). Overt, covert, and subtle sexism. *Psych. Women Q.* 21(1).
- Swim, J. K. et al. (2001). Everyday sexism. *J. Social Issues* 57(1).

Data & Tasks

EXIST 2025 (Plaza et al., 2025)

- 7,958 tweets (EN/ES) + 4,044 memes (EN/ES)
- 887 unique annotators recruited via Prolific
- 6 annotators per instance (3F + 3M), varying across instances

Annotator Pool Demographics

	Tweets (n=809)	Memes (n=887)
Gender	50/50 F/M	50/50 F/M
Age	34% 18-22, 34% 23-45, 32% 46+	33% each band
Education	50% Bachelor's, 31% High school, 15% Master's	Similar
Top countries	Mexico 18%, Portugal 16%, Spain 14%, UK 10%	Spain 13%, Mexico 13%, UK 11%, South Africa 11%
Pool overlap	All 809 tweet annotators also did memes (Jaccard = 0.91)	

Legend: F, 18-22 y.o. (pink), F, 23-45 y.o. (blue), M, 18-22 y.o. (green), M, 23-45 y.o. (purple), M, 46+ y.o. (yellow), F, 46+ y.o. (orange)

Task 1: Detection – “Is this sexist?”

ToM operation: Norm-relevance recognition (*cognitive ToM*)

Example tweets: ID 200712 @tweet200712: go and cook for your husband... you are here tweeting (6/6 YES); ID 201037 @tweet201037: Have you seen The Feminine Mystique? I've lost my copy. (6/6 NO); ID 201037 @tweet201037: He just realized he won't ever understand women (3/6 YES + 3/6 NO)

Task 2: Source Intention – “What is the author's intent?”

ToM operation: Intent attribution (*cognitive ToM*). Distinguishing reporting from condemning requires higher-order ToM (Perner & Wimmer, 1985).

DIRECT: The speaker *perpetrates* sexism
REPORTED: The speaker *reports or describes* a sexist situation
JUDGEMENTAL: The speaker *condemns* sexism

Example tweet: ID 202424 @tweet202424: this is the same little boy who has said 'when we get married, I'm going to slap the shit out of you' (JUDGEMENTAL, REPORTED, DIRECT)

Task 3: Sexism Categorisation – “What is the sexist attitude about?”

ToM operation: Focus identification (mixed *cognitive ToM* + *affective ToM*).

Recognising abstract propositions about gender (*cognitive ToM*)

- IDEOLOGICAL-INEQUALITY:** Rejects gender equality or presents women as inferior
- STEREOTYPING-DOMINANCE:** Assigns stereotypical roles or presents one gender as dominant

Identifying harm directed at persons/bodies (*affective ToM*)

- OBJECTIFICATION:** Reduces a person to their physical appearance or body
- SEXUAL-VIOLENCE:** Describes, threatens, or normalises sexual violence
- MISOGYNY:** Expresses hatred or contempt toward women

Example meme: I LIKE MY WOMEN LIKE I LIKE MY PANCAKES (DIRECT, DIRECT, DIRECT, DIRECT, DIRECT, JUDGEMENTAL)

Example tweet: ID 200643 @tweet200643: \$LUNABROOOO such a cock tease I've been watching this chart all day (OBJECTIFICATION, MISOGYNY, OBJECTIFICATION, STEREOTYPING-DOMINANCE, OBJECTIFICATION, SEXUAL-VIOLENCE, - , IDEOLOGICAL-INEQUALITY)

Research Questions, Predictions & Results

Question	Prediction	Test
P1 Does disagreement increase from detection to interpretation?	Yes, interpretation requires more complex ToM	Entropy + Jaccard across task levels
P2 Does gender structure disagreement?	Only where affective ToM is engaged	Wilcoxon (detection), Fisher's exact (intent, categories)
P3 Which categories are gender-structured?	Affective categories (objectification, misogyny, sexual-violence), not cognitive ones	Per-category Fisher's exact, Holm-corrected

Note on meme detection: Social psychology distinguishes overt from subtle sexism (Swim & Cohen, 1997). Drawing attention to subtle everyday sexism affects women's beliefs more than men's (Becker & Swim, 2011). The small gender effect on meme detection is consistent with this: memes encode sexism through visual framing and irony, i.e., a more subtle, interpretive cue than explicit text.

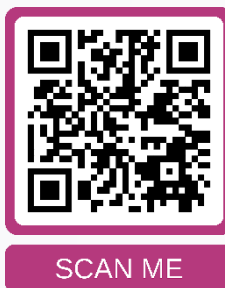
Why misogyny in tweets?
Verbal hostility toward women: women report more everyday sexist incidents and are disproportionately targeted (Swim et al., 2001), producing different impact estimations through experiential proximity.

Why objectification in memes?
Visual/bodily objectification engages the "objectifying gaze" (Mulvey, 1975; Fredrickson & Roberts, 1997). Women's experiential proximity to being objectified produces different focus identification.

Implications

Gender structuring increases with affective ToM demand: negligible for cognitive operations, strongest for affective ones.

- Annotation protocols should separate intent attribution from focus identification
- Perceiver-aware models should condition selectively on annotator identity
- Disagreement is not noise but reflects the ToM structure of social perception



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